

**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**

Examination : Semester Final  
 Duration: **2 Hours**

Semester: **Summer 2025**  
 Full Marks: **70**

CSE421 / EEE465 : Computer Networks

Answer **Sections A, B and C** as per instructions given. (**Pages: 3**)

Figures in the right margin indicate marks.

Name:

ID:

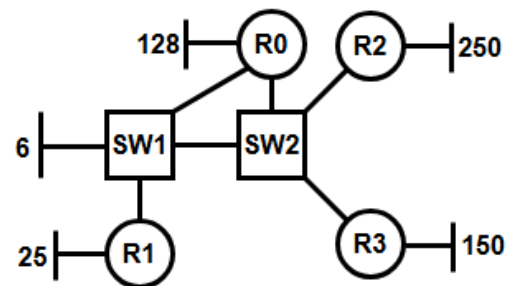
Section:

**SECTION A [All questions of this section are MANDATORY] - 40 MARKS**

**Q1** The number of hosts given in the topology only  
**[CO3]** includes PCs, printers, and servers.

Sheldon and Leonard are both on **R1-LAN**. Their IP addresses are **192.128.0.2** (2nd usable IP address of the network) and **192.128.255.253** (the second last usable IP address of the network), respectively.

- Calculate the prefix mask of the R1-LAN based on the above scenario.
- Howard, who is in R2-LAN decides to send a packet to all the devices on **R1-LAN**. What destination address should Howard use?
- Sheldon now plans to expand his office after winning the Nobel Peace Prize and hence designs the topology given above. **Apply VLSM** using the network address **169.0.0.0/17** to efficiently create the subnet addresses for the expanded topology.
- For the switched network, how many addresses are we wasting?



2  
+  
2  
+  
10  
+  
2

**Q2**

| #   | Data Length | MF  | DF  | Offset |
|-----|-------------|-----|-----|--------|
| ... | ...         | ... | ... | ...    |
| 10  | ...         | 1   | 1   | 1854   |
| ... | ...         | ... | ... | ...    |
| 15  | 69          | 0   | 1   | 2884   |

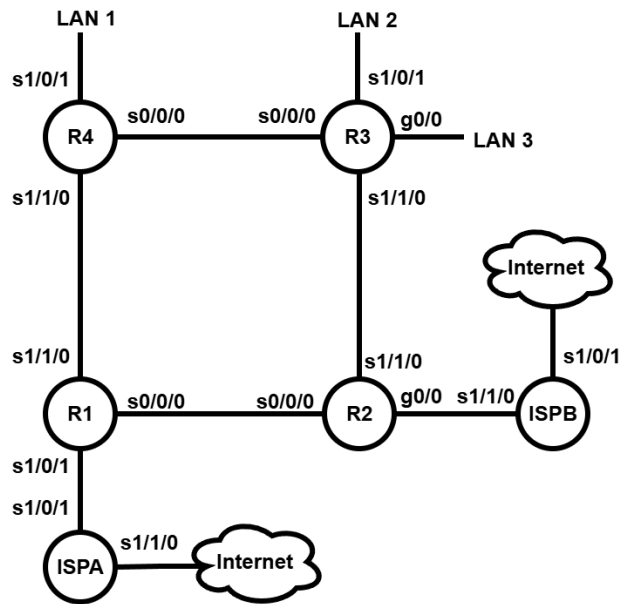
Given, the header size used in this is 38 bytes. From the above table,

- Identify** the packet size of the 10th fragment
- Identify** the data size of the original datagram
- For the offset field, we convert the 16 bit number to 13 bits. Despite shortening it by three bits, how are we preventing any data loss?

4  
+  
3  
+  
2

Q3

| Device | Interface | IP   | Network          |
|--------|-----------|------|------------------|
| R1     | s0/0/0    | .101 | 172.20.10.100/30 |
|        | s1/1/0    | .225 | 172.20.11.224/30 |
|        | s1/0/1    | .1   | 172.20.9.0/30    |
| R2     | s1/1/0    | .193 | 172.20.12.192/30 |
|        | s0/0/0    | .102 | 172.20.10.100/30 |
|        | g0/0      | .1   | 178.51.100.0/30  |
| R3     | s1/0/1    | .1   | 172.20.96.0/24   |
|        | g0/0      | .1   | 172.20.80.0/26   |
|        | s1/1/0    | .194 | 172.20.12.192/30 |
|        | s0/0/0    | .97  | 172.20.10.96/30  |
| R4     | s1/0/1    | .1   | 172.20.64.0/25   |
|        | s0/0/0    | .98  | 172.20.10.96/30  |
|        | s1/1/0    | .226 | 172.20.11.224/30 |
| ISPA   | s1/0/1    | .2   | 172.20.9.0/30    |
|        | s1/1/0    | .1   | 210.1.1.0/24     |
| ISPB   | s1/1/0    | .2   | 178.51.100.0/30  |
|        | s1/0/1    | .1   | 215.1.1.0/24     |

3  
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3  
+  
3  
+  
3  
+  
3

- [CO3] I. On **R4**, configure a **static default route (recursive)** toward the upstream via **R1**.  
 [CO3] II. Configure a **back-up route (directly attached)** of the above static route (AD 5).  
 [CO3] III. On **R3**, add a **recursive** static route to reach the WAN link between **ISPB** and **R2**.  
 [CO2] IV. What inefficiency will the static route in (III) create?  
 [CO2] V. Two default static routes are configured on **R2**: S\* 0.0.0.0/0 [10/0] via 178.51.100.2 and S\* 0.0.0.0/0 [20/0] via 172.20.10.101. Which entry should be in the routing table of R2, and why?

### END OF SECTION A

[CO3] **SECTION B** [Answer ANY TWO out of THREE in this section] - 12 MARKS

- Q4** Assume an OSPF-like **link state** (on the topology mentioned in **Q3**) inside the four-router area {R1, R2, R3, R4}. **ISPA/ISPB** are external (no LSPs from them in the area unless stated).

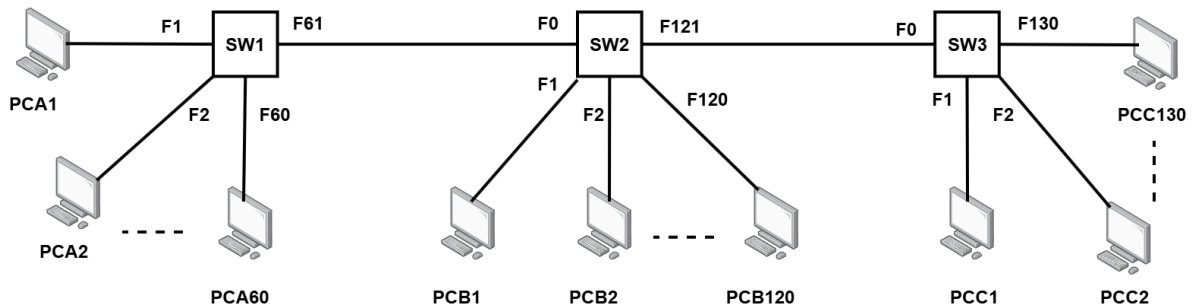
4  
+  
2

- I. The **R3–R2** link goes down. R3 creates one new LSP. **Identify** how this LSP packet differs from the initial LSP packet that was created by R3 at the beginning of Link State Routing, and to which interfaces R3 will forward this packet.  
 II. If **Hellos** between **R1** and **R4** stop (timer expires), what happens in the link-state?

- Q5** I. **Apply** your knowledge of IPv6 address formatting to transform and **simplify** the following addresses into their expanded form according to IPv6 rules.  
 A. ::1:0:0  
 B. fe80:2ac:66::0:1000:100:10  
 C. For (B), identify the type of address.  
 II. IPv6 introduces a new type of address called Anycast. **Explain** with an example how Anycast is useful on the Internet.

1  
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1  
+  
1  
+  
3

Q6

3  
+  
3

Consider the above switched network. The host PCB1, connected to SW2 sends a frame to PCC2, connected to SW3. All switch tables are empty **except** SW3, which has the information of PCC2 in its table.

- I. How many PCs will receive the request frame? Why?
- II. Draw the switch tables of all 3 switches once the reply is received by PCB1.

### END OF SECTION B

[CO2] SECTION C [Answer **ANY THREE out of FIVE** in this section] - 18 MARKS

Q7

- I. How does a device receiving an ICMP packet know that it is a PING Echo request?
- II. When running traceroute, why do you sometimes see \* \* \* instead of a router's reply?
- III. State one key difference between a datagram network and a virtual circuit network.

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2  
+  
2

Q8

A laptop is assigned an IP address by DHCP with an 8-hour lease. After 4 hours of active use, it tries to renew the lease.

- I. Which message does the laptop send to attempt renewal?
- II. What happens if the server is available and responds?
- III. What fallback mechanism occurs if the server does not respond?

2  
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2  
+  
2

Q9

Two PCs in a LAN (192.168.10.10 and 192.168.10.11) connect at the same time to different web servers on the Internet. The ISP gives only one public IPv4 address (209.165.200.225).

- I. How can both PCs use the Internet with just one public IP?
- II. How does the router know which reply belongs to which PC?
- III. What happens if the same port number is already in use?

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2  
+  
2

Q10

Some devices in a network only support IPv4, while others only support IPv6. To avoid communication problems, the network administrator configures all devices to support both IPv4 and IPv6 simultaneously.

- I. What is this networking approach called?
- II. Why is it considered the most flexible during the transition from IPv4 to IPv6?
- III. What challenge may arise if this system has to be maintained for thousands of devices worldwide?

2  
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2  
+  
2

Q11

A PC (192.168.10.50) prints to its network printer (192.168.10.200, MAC:

3

0C-43-00-12-BC-DD). The printer is replaced with a new one using the same IP address. Initially, the PC cannot print, but after approximately 10 minutes, printing works automatically.

+  
3

- I. Why could not the PC print immediately after the printer was replaced?
- II. What could have been done to enable printing immediately after the printer replacement?

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**END OF SECTION C**

===== **THE END** =====

*Why did the packet sit alone at lunch?*

*It didn't know how to broadcast itself.*